TPOV-Seg: Textually Enhanced Prompt Tuning of Vision-Language Models for Open-Vocabulary Remote Sensing Semantic Segmentation

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***Abstract*—The remote sensing semantic segmentation faces significant challenges in open-world scenarios due to domain gaps and the presence of unseen categories in the testing datasets. Open-vocabulary semantic segmentation (OVSS) based on vision-language models (VLMs) has emerged as a promising paradigm for remote sensing imagery interpretation, which enables adaptation to new datasets with arbitrary semantic categories. However, current OVSS approaches often struggle to achieve fine-grained pixel-level localization and classification for unseen categories when relying solely on fixed textual prompts and pretrained VLM encoders. The model's generalization capability is further hindered by insufficiently fine-grained and adaptive textual representations. To address these limitations, we propose TPOV-Seg, a textually enhanced prompt tuning for OVSS (TPOV-Seg). Specifically, a remote sensing-specific Text TempLator (TTL) is introduced to enrich textual prompts and semantic representations for land cover categories by incorporating synonymous vocabulary combinations. To efficiently align the text encoder with remote sensing characteristics, a lightweight text-aware prompt tuning (LTP-Tuning) strategy is proposed for contextual modeling of word embedding adaptation. Furthermore, a textual-guided channel-aware aggregator (TGCA) is developed to promote interchannel feature interaction and facilitate semantic modeling, leveraging grouped cross-channel Transformers and linear Transformers under the guidance of enhanced textual features from TTL. Extensive experiments on five large-scale remote sensing segmentation datasets demonstrate that TPOV-Seg outperforms existing methods in OVSS tasks, showing strong discriminative ability for unseen categories while maintaining the robust cross-domain generalization. The source codes will be available at: https://github.com/zxk688/TPOVSeg**

***Index Terms*—Open vocabulary, prompt tuning, remote sensing, semantic segmentation, zero shot.**

I. Introduction

REMOTE sensing semantic segmentation is a fundamental task in Earth's observation research and applications, such as natural resource management [1], urban planning [2], disaster monitoring [3], and environmental protection [4]. It is essentially a pixelwise classification task on remote sensing images, aiming to assign each pixel to a specific land cover category [5], [6]. In recent years, deep learning techniques have significantly improved the semantic segmentation accuracy on large-scale remote sensing datasets by learning high-level abstract representations directly from high-resolution image data, making it a core approach for Earth's observation applications [7], [8]. Despite the achievements, the remote sensing semantic segmentation performance is still hindered by the reliance on a large number of training datasets and weak generalization capability on heterogeneous datasets [9]. To address this issue, developing transferable models with cross-domain generalization capability has attracted sustained research attention [10], [11]. To achieve this, one line of research focuses on the cross-domain semantic segmentation, which leverages unsupervised domain adaptation (UDA) techniques to transfer models trained on a labeled source domain to an unlabeled target domain without requiring annotations in the latter [9], [12]. However, a key limitation is that the training process still relies on access to unlabeled data from the target domain to achieve the domain feature alignment. Moreover, domain shifts can only be mitigated when the two domains exhibit moderate distribution discrepancies and share a consistent label space [13], [14], [15]. However, these conditions are not always met in real-world applications, where statistical heterogeneity and semantic shifts often coexist. As a result, models trained using UDA still struggle to rapidly recognize unseen categories and adapt to domain shifts that involve semantic discrepancies [16]. On the other hand, the pretraining–fine-tuning paradigm has been applied in the remote sensing semantic segmentation by transferring the models pretrained on a large-scale dataset to a downstream task with limited labeled samples. These models pretrained in a self-supervised [17], [18], [19] or a fully supervised manner [20] can provide a better initialization for the parameter updates of the downstream semantic segmentation network and facilitate the transfer learning. However, the pretrained models only provide the image encoder part with extra modules required to construct the semantic segmentation networks, which still rely on labeled samples for fine-tuning or retraining.

Recently, there has been a growing consensus that the development of generic and scalable remote sensing semantic segmentation approaches is essential and more

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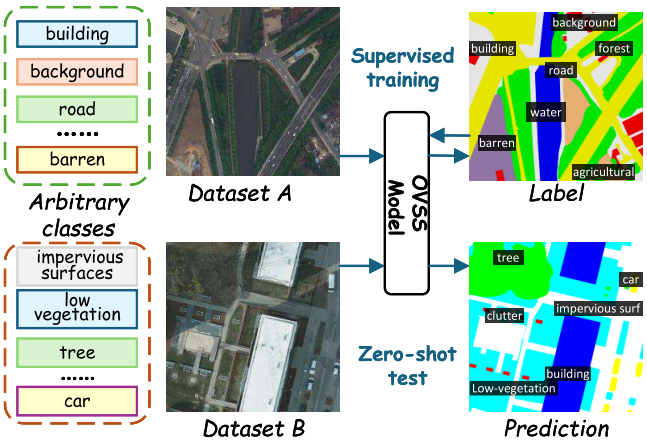


Fig. 1. Illustrations of the training and testing processes of OVSS.

practical for real-world scenarios [21], [22]. Therefore, the open-vocabulary semantic segmentation (OVSS) has become a popular paradigm for remote sensing image interpretation [23], [24]. By utilizing pretrained vision-language models (VLMs), like contrastive language-vision pretraining (CLIP) [25], it enables the handling of a much wider variety of land cover classes without needing explicit training data for each new category to achieve zero-shot segmentation performance [26], [27], [28]. Specifically, it attempts to recognize and segment arbitrary categories, even previously unseen categories, using textual descriptions or semantic knowledge beyond the training set. Leveraging cross-modal fusion of text and visual features obtained by CLIP, it enhances the model's ability to distinguish land cover classes [29], [30]. As shown in Fig. 1, OVSS models are normally trained on a large-scale labeled dataset and tested on other datasets with arbitrary semantic categories. The main challenge of OVSS lies in achieving fine-grained pixel-level localization and classification for unseen categories in the absence of pixelwise supervision. The model must rely solely on textual semantics to generalize to novel classes, which places high demands on vision-language alignment, semantic generalization capability, and contextual representation learning.

Since OVSS is a relatively new setting in both the computer vision and remote sensing communities, only limited research has been conducted in this area. Existing works can be broadly categorized into two groups: visual feature enhancement and image–text feature alignment optimization. For visual feature enhancement, representative approaches include introducing an additional image encoder adapted from the segment anything model (SAM) [23], employing a side adapter network attached to the pretrained CLIP model [27], and designing a dual-branch feature fusion strategy [31]. In terms of image–text alignment optimization, recent works have introduced a rotation-aggregative similarity computational module [24] and a spatial-class cost aggregation approach [32]. Different from previous methods, we focus on introducing more diverse semantics of land cover categories to enhance the textual representation capability and the cross-domain generalization.

In OVSS, the transferability of learned visual–textual mappings may degrade due to significant domain shifts caused by statistical heterogeneity and semantic shifts [33]. Predefined or manually crafted prompts often fail to capture the broad spectrum of potential object descriptions due to the limited diversity and adaptability of prompt designs, especially when category naming inconsistencies exist across different datasets. We argue that an overemphasis on visual features may still result in the suboptimal alignment with the textual content, due to insufficient fine-grained textual representations required for precisely associating linguistic cues with corresponding visual elements. This misalignment can hinder the model's ability to generalize effectively to novel or unseen concepts.

To address this gap, we propose a novel textually enhanced prompt tuning for OVSS (TPOV-Seg) on remote sensing images. Specifically, a pool of textual prompt templates with multiple augmentations is constructed to generate diverse prompts for each category. Consequently, the pretrained CLIP textual representations are obtained by aggregating features from multiple prompt templates, enabling the model to learn more diverse textual representations and context-rich semantics from the CLIP text encoder. We further enhance the text encoder by fine-tuning it with lightweight text-aware learnable prompts. This design unleashes the extensive knowledge embedded in CLIP through large-scale pretraining, while preserving its ability to capture domain-invariant properties across diverse remote sensing scenarios. To aggregate the multimodal cost volume—i.e., the cosine similarity scores between image and text embeddings—a novel aggregator is proposed that incorporates channel, class, and spatial properties under textual guidance to better capture the spatial structure and semantic context of objects. The main contributions are summarized as follows.

1) A remote sensing-specific Text TempLator (TTL) that leverages synonymous vocabulary combinations to fully exploit category-related textual knowledge and enhance the model's textual representation and semantic understanding capability across different domains. Furthermore, a lightweight text-aware prompt tuning (LTP-Tuning) approach for contextual word embeddings is introduced to achieve effective adaptation of the text encoder to remote sensing characteristics.
2) A textual-guided channel-aware aggregator (TGCA) is developed to prioritize the aggregation of the cost volume along the channel dimension based on grouped cross-channel Transformer and linear Transformer, using the augmented textual features generated by TTL as guidance. By promoting cross-channel feature interaction, the network facilitates more effective modeling of semantic relations and spatial structures.
3) Extensive experiments on five large-scale remote sensing datasets demonstrate the effectiveness of TPOV-Seg for OVSS, showing strong generalization performance across heterogeneous datasets and an improved ability to capture heterogeneous semantics.

The remainder of this article is structured in the following manner. Section II provides a concise review of related work on transfer learning and OVSS. Section III outlines the key

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components of the proposed TPOV-Seg. Section IV presents the performance evaluation and comparative analysis of TPOV-Seg. Finally, Section V concludes this article.

II. Related Works

A. Remote Sensing Semantic Segmentation

Over the years, researchers have explored various network architectures like fully convolutional networks (FCNs), U-Net, Vision Transformer (ViT), and Swin Transformer for remote sensing semantic segmentation tasks [34], [35], [36]. Specifically, FCNs and U-Net pioneered the use of end-to-end convolutional neural networks (CNNs) with an encoder–decoder architecture for pixelwise classification. Building on this foundation, ViTs have emerged as the dominant paradigm, largely due to their superior ability to model long-range dependencies and capture global contextual information [37], [38]. However, in high-resolution remote sensing images, the significant scale variation and intricate object details require not only robust global contextual information but also the preservation of fine-grained local features for accurate boundary delineation. Consequently, a central focus of semantic segmentation research has been the effective modeling of global–local representations from ViTs [39].

However, these models are typically trained in a fully supervised manner and rely on a large amount of specific annotated samples. Due to the varying acquisition conditions and image spectral shifts, semantic segmentation networks trained on specific annotated data struggle to achieve promising generalization performance on heterogeneous remote sensing image datasets.

B. Transferable Semantic Segmentation Models

Training a transferable model for cross-domain semantic segmentation has drawn sustainable attention in remote sensing [9], [11]. To train transferable semantic segmentation models, two main lines of research have been explored: UDA and fine-tuning of pretrained models.

UDA techniques aim to transfer knowledge from a labeled source domain to an unlabeled target domain using samples from two domains for collaborative training. Overall, the cross-domain alignment for semantic segmentation can be achieved by adversarial training [2], [15], [40], [41], [42] or pseudo-labeling [11], [43], [44]. However, these methods rely on two assumptions that do not always hold in real-world applications: 1) the source and target domains share the same label space and 2) the target data should be available during training for domain-invariant feature learning. As a result, they often struggle with unseen domains containing new classes absent in the source domain or out-of-distribution samples that deviate from the source data distribution.

Recent remote sensing pretrained models have advanced the remote sensing interpretation by introducing general domain knowledge from large-scale remote sensing datasets. These models are typically pretrained on large-scale unlabeled remote sensing data using self-supervised learning to acquire general image feature representations [45]. Alternatively, the pretrained models can be built using publicly available labeled datasets [20]. As a result, they enhance the adaptability of pretrained models through transfer learning, provide semantic segmentation networks with proper parameter initialization, and facilitate more efficient network parameter optimization. By using pretrained encoders and loading the weights, these models can be easily transferred to various downstream tasks across different remote sensing datasets [20], [46]. Recent studies have investigated the adaptation of vision foundation models, i.e., SAM, for remote sensing semantic segmentation [47], [48]. For instance, MeSAM [48] incorporated inception Transformers and multiscale connections to effectively address scale variations in objects within remote sensing imagery. However, for the downstream semantic segmentation tasks, pretrained foundation models still rely on labeled samples either to fine-tune the entire network or to apply parameter-efficient fine-tuning strategies that keep the pretrained encoder frozen [49], [50].

C. VLMs-Based OVSS

Recently, vision-language foundation models have significantly impacted the field of computer vision and remote sensing [19], [51]. A notable advancement is CLIP, which effectively bridges the gap between images and natural language by training on large-scale data within a multimodal embedding space. CLIP has demonstrated strong transfer learning capabilities, achieving zero-shot performance on the image classification task.

To leverage CLIP's powerful semantic knowledge for OVSS, OpenSeg [52] proposed to use image captions as supervision to learns visual-semantic alignments in a weakly supervised learning manner. Moreover, OVSeg [26] adopted a two-stage approach, where class-agnostic mask proposals are first generated, followed by mask classification using the pretrained CLIP model. In comparison, MaskCLIP [53] developed a Transformer-based class-agnostic visual encoder to improve the original CLIP visual encoder. In addition, side adapter network (SAN) [27] introduced an additional side adapter network to predict mask proposals and utilized the frozen CLIP to recognize the classes. Furthermore, CAT-Seg [32] focused on the matching costs between image and text embeddings and aggregated the costs at the spatial and class level. These methods primarily address the core challenge of OVSS: generating high-quality mask proposals and classifying them accurately using a VLM's vast vocabulary knowledge. In comparison with natural scene images, high-resolution remote sensing images display greater variability in object appearances and environmental conditions across different regions. Furthermore, land cover/use category names differ from those in natural image datasets, and the same category may be described or defined inconsistently across different remote sensing datasets.

OVSS has recently emerged as a prominent research direction in remote sensing, enabling segmentation beyond fixed category sets and motivating the development of specialized VLM-based approaches. For instance, Ye et al. [31] proposed a dual-stream image encoder for specialist and generalist feature extraction with a feature fusion module to integrate them. In [24], a rotation-aggregative similarity computational

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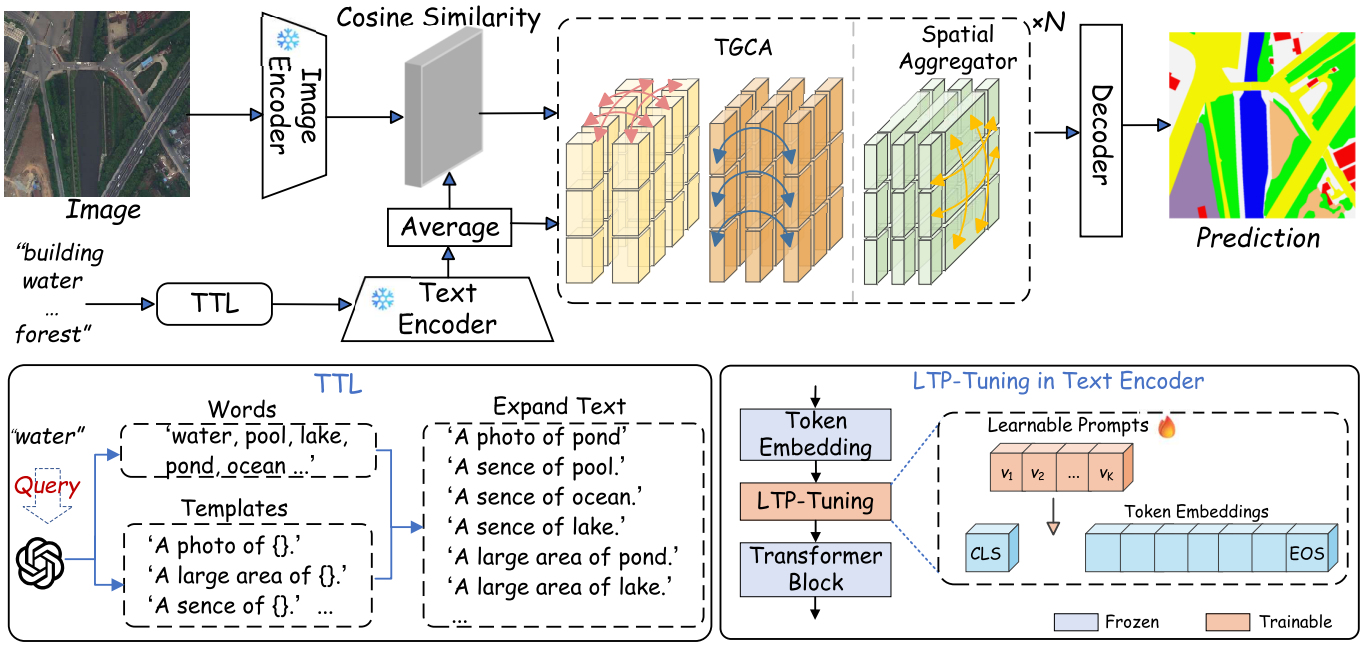


Fig. 2. Framework of the proposed TPOV-Seg. Specifically, a remote sensing-specific TTL is designed to construct a diverse pool of textual prompt templates with synonym-based augmentation and enables the CLIP text encoder to generate rich and context-aware textual representations. To efficiently adapt the text encoder to remote sensing characteristics, we introduce the LTP-Tuning strategy. In addition, the aggregator containing TGCA and the spatial aggregator is developed to aggregate the multimodal cost volume by jointly modeling channelwise and spatial interactions under textual guidance and enhancing both semantic discrimination and spatial structure understanding.

module is introduced to produce orientation-adaptive similarity maps, which are utilized as the initial semantic representations. AerOSeg [23] introduced an additional image encoder adapted from SAM to provide complementary image features. To reduce the training cost, SegEarth-OV [28] utilized a universal upsampler with a simple subtraction operation to reduce bias in the patch token representations using only unlabeled images.

Overall, these open-vocabulary methods primarily focus on visual feature enhancement and image–text feature alignment optimization at the pixel level to adapt global VLM knowledge for dense prediction. However, a major limitation remains the reliance on fixed VLM features and the common use of manually designed category names as prompts, which often fail to capture diverse textual features and restrict adaptability to novel domains.

III. Methodology

As shown in Fig. 2, our framework consists of a CLIP image encoder, a CLIP text encoder with TTL and LTP-Tuning, an aggregator, and a decoder. Remote sensing images are first processed by the CLIP image encoder to extract visual features. Meanwhile, the CLIP text encoder encodes corresponding textual descriptions into semantic embeddings. Then, the multimodal cost volume is computed by cosine similarity. After that, a novel aggregator considering channel, class, and spatial properties is proposed to refine the cost volume, while the decoder reconstructs the pixelwise classification results.

A. Preliminaries

Let the input high-resolution remote sensing image be represented as $I \in \mathbb{R}^{H\times W\times 3}$, where H and W represent the spatial dimensions and the three channels correspond to the RGB bands. Normally, the CLIP text encoder takes as input a label set $\mathcal{C} = \{\tau_c\}_{c=1}^{C}$, where each τ_c represents the textual expression associated with the cth class, and C is the total number of categories. For example, a possible form of τ_c might be "an image of a [CLASS]." The objective of OVSS is to predict a category for each pixel in I from the label set $\mathcal{C}$. Unlike conventional segmentation tasks with fixed and predefined classes, this setting involves a more general and adaptable category set described by natural language, posing unique challenges in model generalization and flexibility, i.e., $\mathcal{C}_{\text{test}} \setminus \mathcal{C}_{\text{train}} \neq \emptyset$. Following CLIP, the vision encoder $g^V(\cdot)$ and text encoder $g^L(\cdot)$ are adopted for extracting image and text embeddings, respectively.

B. Text TempLator

The TTL serves as a text template generator in our framework, designed to address the limitations of traditional single-prompt approaches in zero-shot segmentation. By leveraging a combination of candidate words and predefined templates, TTL ensures that each category is represented by a rich set of linguistic expressions and enables the model to capture the nuanced relationships between text and visual content. To generate diverse and fine-grained vocabulary expansions for each land cover category, we use OpenAI's GPT-4.5 by prompting as follows.

Q:Given the keyword [CLASS] used in remote sensing image semantic segmentation, generate the following.

1) Semantically related words from multiple perspectives (types, states, attributes, and contexts). For example, for "water": ocean, lake, river, ice, and clear water.

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2) Diverse natural language templates with {} as placeholders. For example, "a photo of {}" and "a large area of {}."

After that, we refine TTL by manually filtering out irrelevant or redundant phrases to ensure the semantic consistency and diversity. Given a set of candidate words (such as "water," "river," and "lake") and predefined templates (such as "a photo of {}," "a landscape with {}," and "a large area of {}"), TTL expands each word with all templates to generate a comprehensive set of prompts. Formally, let $\mathcal{W} = \{w_1, w_2, \ldots, w_{M_1}\}$ be the word set and $\mathcal{T} = \{\tau_1, \tau_2, \ldots, \tau_{M_2}\}$ represent the template set, where M_1 and M_2 are the numbers of words and templates, respectively. The expansion process can be expressed as follows:

$$\mathrm{TTL}(\mathcal{W}, \mathcal{T}) = \left\{\tau_j(w_i) | w_i \in \mathcal{W}, \tau_j \in \mathcal{T}\right\} \tag{1}$$

where $\tau_j(w_i)$ represents the filling word w_i into template τ_j. For example, "water" with the template "a photograph of [CLASS]" generates "a photograph of water." This mechanism systematically creates a diverse set of language descriptions, enabling richer text–image alignment learning. This template expansion approach offers several significant advantages over single-prompt methods. First, it substantially increases semantic space coverage by introducing various contexts and expressions, allowing the model to understand the same concept across different linguistic settings. For instance, "river" can be described as "a winding river," "a river area," or "river landscape," each capturing different characteristics and visual representations of rivers. Second, the diverse prompt set generated by TTL helps to reduce the model's dependencies on specific expressions, improving robustness when facing novel descriptions or uncommon phrasings.

C. Lightweight Text-Aware Prompt Tuning

The LTP-Tuning uses learnable vectors for context modeling to bridge the imbalance between fixed text descriptions and the rich information at the image level. To achieve this, we introduce learnable prompt tokens [54], [55] that can adapt the pretrained CLIP model to the specificities of the remote sensing domain.

Given a text embedding $x_{i,j} \in \mathbb{R}^{S\times D}$ acquired by $\tau_j(w_i)$, where S is the text sequence length, and D is the embedding dimension. We define the learnable prompt vectors as $v \in \mathbb{R}^{K\times D}$, where K is the length of the prompt vectors. The prompt tokens are inserted into the embedded sequence through the prompt insertion function $\mathcal{P}(x)$

$$\mathcal{P}(x) = [x[0:1], v, x[1:]] \in \mathbb{R}^{(S+K)\times D} \tag{2}$$

where $x[0:1]$ represents the embedding of the first token (e.g., [CLASS]), v represents the learnable prompt tokens, and $x[1:]$ denotes the remaining token embeddings. The prompt tokens are initialized as zero tensors and converted to learnable parameters using half-precision (float16) format. After that, the augmented sequence is fed into the Transformer-based text encoder

$$\mathbf{h}_{i,j}^{L} = g^{L}\left(\mathcal{P}\left(x_{i,j}\right)\right) \in \mathbb{R}^{1\times D} \tag{3}$$

where $g^L(\cdot)$ is the text encoder that maintains the input dimension in its output representation and $\mathbf{h}^L$ represents the token embeddings with prompts encoded after passing through 12 Transformer layers.

As illustrated in Fig. 3, the CLIP text encoder consists of 12 Transformer layers. Learnable prompt vectors are inserted into the token embeddings before entering the Transformer layers and removed after the attention computation is completed. This strategy offers several key advantages. First, since the prompt vectors are only temporarily inserted during attention computation and removed thereafter, the original model structure and sequence length remain unchanged across layers, avoiding issues caused by cumulative modifications. Second, the method is parameter-efficient, requiring only a small number of learnable parameters (i.e., embedding vectors), without the need to fine-tune the entire pretrained model, thereby significantly reducing computational and storage costs. Finally, by keeping the pretrained model parameters frozen, it effectively mitigates catastrophic forgetting, reducing the risk of the model overfitting to new tasks and forgetting previously acquired knowledge.

D. Aggregator

Normally, the similarity computation takes image feature embeddings $\mathbf{h}^V \in \mathbb{R}^{H/4\times W/4\times D} = g^L(I)$ extracted from the CLIP image encoder and text features generated from the CLIP text encoder as input. In comparison, we use the average feature embeddings $\bar{\mathbf{h}}^L \in \mathbb{R}^{C\times D}$ obtained by various text templates in TTL as text embeddings

$$\bar{\mathbf{h}}^L[c] = \frac{1}{M_1 M_2}\sum_{i=1}^{M_1}\sum_{j=1}^{M_2}\mathbf{h}_{i,j}^{L} \in \mathbb{R}^{1\times D} \tag{4}$$

$$\bar{\mathbf{h}}^L = \left[\bar{\mathbf{h}}^L[1];\ \bar{\mathbf{h}}^L[2], \ldots, \bar{\mathbf{h}}^L[C]\right] \in \mathbb{R}^{C\times D}. \tag{5}$$

For each spatial position in the image feature maps and each category in the candidate set, we can compute the cosine similarity as a cost volume between them, i.e., a tensor consisting of the similarities between each pixel in the image and the text embeddings of all categories [32], as follows:

$$\Phi[q, c] = \frac{\mathbf{h}^V[q]\cdot\bar{\mathbf{h}}^L[c]}{\left\|\mathbf{h}^V[q]\right\|\ \left\|\bar{\mathbf{h}}^L[c]\right\|} \tag{6}$$

where q indexes the 2-D spatial positions of the image embedding. After that, the TGCA operates on the cost volume to produce aggregated feature representations that encode contextual dependencies and category-aware cues, while spatial aggregation is introduced to obtain smooth and continuous object regions. Notably, the visual embeddings and text embeddings extracted from the CLIP encoder are used in the attention calculation during spatial aggregation and TGCA processes, respectively, providing rich semantic and textual features as guidance.

1) Textual-Guided Channel-Aware Aggregator: Remote sensing imagery typically exhibits substantial distributional differences from natural scenes. To further enhance the framework's adaptability to domain-specific characteristics of remote sensing imagery, we introduce the TGCA module to

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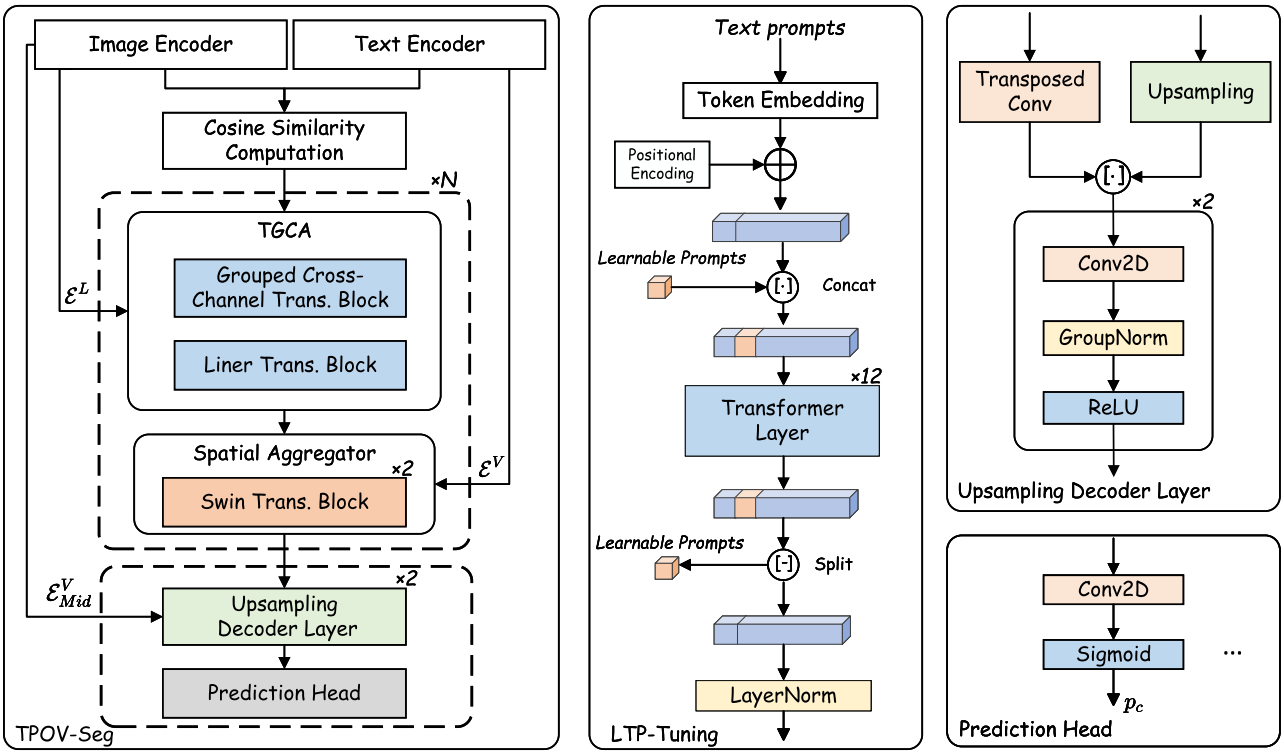


Fig. 3. Network structure of TPOV-Seg.

capture the unique channel expression patterns within the image–text cost volume. It enables free information flow across channel dimensions, where the augmented text generated by TTL for remote sensing imagery serves as encoded guidance. This design facilitates adaptive learning of channel importance by selectively enhancing informative channel features and suppressing noise, thereby improving the semantic representation quality. By introducing channel-level adaptation as the initial processing stage, the framework can facilitate the subsequent learning of interclass relationships and spatial patterns. Specifically, TGCA consists of a grouped cross-channel Transformer block (GCTB) and a linear Transformer block (LTB).

Given a processed cost slice $\tilde{\Phi}[q,:] \in \mathbb{R}^{1\times C\times d_F}$ after a convolutional layer and the guidance feature $\mathcal{E}^L = f_{\text{proj}}(\bar{\mathbf{h}}^L) \in \mathbb{R}^{1\times C\times d_F}$ from CLIP language encoder, the channel-aware aggregation first computes adaptive class weights through a cross-attention mechanism

$$Q = W_Q\left[\tilde{\Phi}\left[q,:\right];\mathcal{E}^L\right], K = W_K\left[\tilde{\Phi}\left[q,:\right];\mathcal{E}^L\right]$$
$$V = W_V\tilde{\Phi}\left[q,:\right] \tag{7}$$

$$\mathbf{w}_{\text{att}} = \text{softmax}\left(\frac{\text{QK}^\top}{\sqrt{d_F}}\right)V \tag{8}$$

where $\mathbf{w}_{\text{att}}$ represents the adaptive class weights that capture the importance of different channels for each class. As shown in Fig. 4, query and key are generated from the concatenated feature, while the value comes from the original visual feature, where W_Q, W_K, and W_V are learnable projection matrices. The generated weights are then applied to the original cost volume

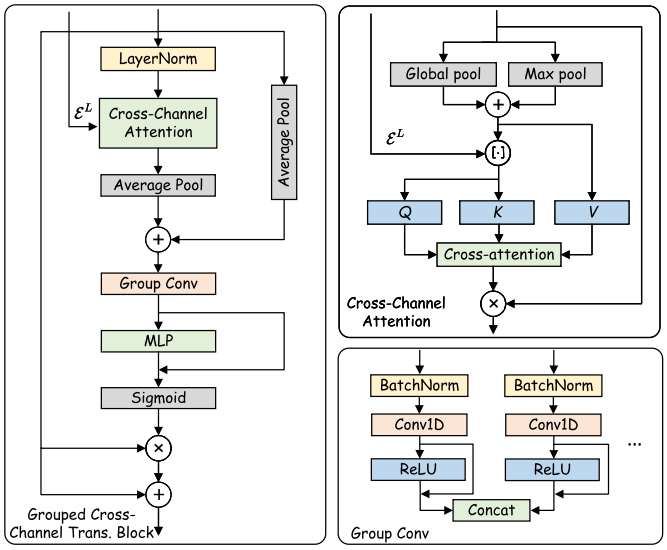


Fig. 4. Illustration of the details of the proposed GCTB.

through a weighted connection

$$\tilde{\Phi}_{\text{att}}\left[q,:\right] = \tilde{\Phi}\left[q,:\right] \odot (\mathbf{1} \oplus \mathbf{w}_{\text{att}}) \tag{9}$$

where $\odot$ represents the elementwise multiplication and $\oplus$ denotes the elementwise addition operation. The attention-weighted features $\tilde{\Phi}_{\text{att}}$ are subsequently fed into a group convolutional module to capture interchannel dependencies, which can be expressed by

$$\mathbf{w}_{\text{group}} = \text{GroupConv}\left(\text{AvgPool}\left(\tilde{\Phi}_{\text{att}}\left[q,:\right]\right)\right) \tag{10}$$

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where GroupConv is used to process channel subsets independently and facilitate structured and efficient channelwise interaction. After that, $\mathbf{w}_{\text{group}}$ is combined with an MLP layer followed by Sigmoid activation to generate the final channel weights $\mathbf{w}_{\text{ch}}$

$$\mathbf{w}_{\text{ch}} = \sigma\left(\mathbf{w}_{\text{group}} \oplus \text{MLP}\left(\mathbf{w}_{\text{group}}\right)\right) \tag{11}$$

where σ means the Sigmoid function. The final channel weights $\mathbf{w}_{\text{channel}}$ are applied to guide the optimal utilization of features at different channels

$$\tilde{\Phi}_{\text{ch}}\left[q,:\right] = \tilde{\Phi}\left[q,:\right] \odot \left(\mathbf{1} \oplus \mathbf{w}_{\text{ch}}\right). \tag{12}$$

This channel-aware processing enables the framework to adaptively weight different channel features, providing optimal guidance for subsequent class aggregation modules. Unlike standard channel attention mechanisms such as squeeze-and-excitation (SE) attention [56] or efficient channel attention (ECA) [57], which primarily capture channelwise dependencies through the global pooling and a lightweight gating function, the proposed TGCA introduces a cross-channel attention mechanism guided by enhanced textual features. Specifically, TGCA computes adaptive class weights by jointly modeling the visual cost volume and the augmented textual embeddings, enabling the semantic-aware channel reweighting rather than relying solely on statistical pooling. This design allows free information flow across channels under textual guidance, selectively enhancing discriminative features while suppressing noise. In contrast, SE and ECA lack this modality-aligned guidance, resulting in limited adaptability to domain-specific semantics in remote sensing imagery.

After that, the class aggregation module processes spatial features to capture interclass relationships as follows:

$$\Phi_{\text{ca}}\left[q,:\right] = \mathcal{A}^{\text{LTB}}\left(\left[\Phi_{\text{ch}}\left[q,:\right];\mathcal{E}^{L}\right]\right), \quad \Phi_{\text{ch}}\left[q,:\right] \in \mathbb{R}^{C\times d_F} \tag{13}$$

where $\Phi_{\text{ca}}[q,:]$ represents the output features after class aggregation at the spatial position q. Furthermore, $\mathcal{A}^{\text{LTB}}$ represents LTB without positional embedding to ensure permutation invariance across classes. This design enables the model to handle varying numbers of classes during inference while maintaining effectiveness for both seen and unseen categories. As a result, the semantic relationships between different categories are integrated so that the model can more accurately distinguish or associate categories with similar semantics.

2) Spatial Aggregator: The spatial aggregation is the process of introducing spatial structural information of the image, such as smoothness and continuity, during the alignment of image and text features, to enhance the model's ability to make spatially coherent predictions. As a result, it can maintain smoothness within the image by processing each class independently, which can be expressed as follows:

$$\Phi_{\text{sa}}\left[:,c\right] = \mathcal{A}^{\text{STB}}\left(\left[\Phi_{\text{ca}}\left[:,c\right];\mathcal{E}^{V}\right]\right), \quad \Phi_{\text{ca}}\left[:,c\right] \in \mathbb{R}^{\left(\frac{H}{16}\times\frac{H}{16}\right)\times d_F} \tag{14}$$

where $\Phi_{\text{ca}}[:,c]$ represents the feature map of the cth category, $\Phi_{\text{sa}}[:,c]$ is the spatially aggregated result, and $\mathcal{E}^{V} = f_{\text{proj}}(\mathbf{h}^{V})$ means the projected image feature embeddings. $\mathcal{A}^{\text{STB}}$ applies a pair of consecutive Swin-Transformer blocks (STBs) operating on local and shifted windows.

Overall, the TGCA module captures semantic relationships between different categories and enhances the model's open-vocabulary semantic understanding. The spatial aggregation preserves local consistency within the image. Through iterative refinement with multiple units, the model progressively enhances feature representations to achieve high-quality semantic segmentation.

E. Decoder

The multistage transposed convolution-based decoder is shown in Fig. 3, which consists of two upsampling decoder layers and a prediction head.

In the upsampling decoder layer, the visual embeddings $\mathcal{E}^{V}_{\text{Mid}}$ from the middle layers of the CLIP image encoder undergo transposed convolutions to generate an upsampled feature map, and the features output by the aggregator are upsampled and fused with the feature map. Then, the fused representation is refined through two consecutive processing units, each comprising three sequential operations: a 2-D convolution (Conv2D), followed by group normalization (GroupNorm) that standardizes feature distributions to stabilize the training process, and a rectified linear unit (ReLU) activation function.

Finally, the output is generated through the prediction head that adopts a concise and effective design, receiving refined feature maps from the decoder as input. It first processes the features through a convolutional layer that maintains spatial resolution while modifying channel depth to match the target class count. Subsequently, a Sigmoid activation function maps each pixel's output to the probability of the corresponding class. The entire prediction head structure is simple yet powerful, efficiently transforming the features extracted by the decoder into the final segmentation results.

F. Training and Inference

We use binary cross-entropy (BCE) loss to train our model, updating only the learnable tokens in LTP-Tuning, the aggregator, and the decoder while keeping the encoder frozen. The loss is defined as follows:

$$\mathcal{L} = -\frac{1}{C}\sum_{c=1}^{C}\left[y_c \log\left(p_c\right) + \left(1-y_c\right)\log\left(1-p_c\right)\right] \tag{15}$$

where p_c denotes the predicted probability for class c, y_c represents the corresponding ground truth, and C is the total number of classes. This formulation treats the multiclass segmentation problem as C separate the binary classification problems, enabling the model to dynamically handle datasets with various categories.

During inference, the model performs OVSS by accepting arbitrary textual class descriptions as input to generate pixel-level predictions for the input image. The CLIP text encoder processes these free-form descriptions to produce corresponding text embeddings, which are then compared with dense image features via cosine similarity to compute pixel-level affinities between visual and textual embeddings. The resulting high-resolution features after the aggregator and the decoder enable pixelwise classification for any specified vocabulary without requiring model retraining, as the open-vocabulary

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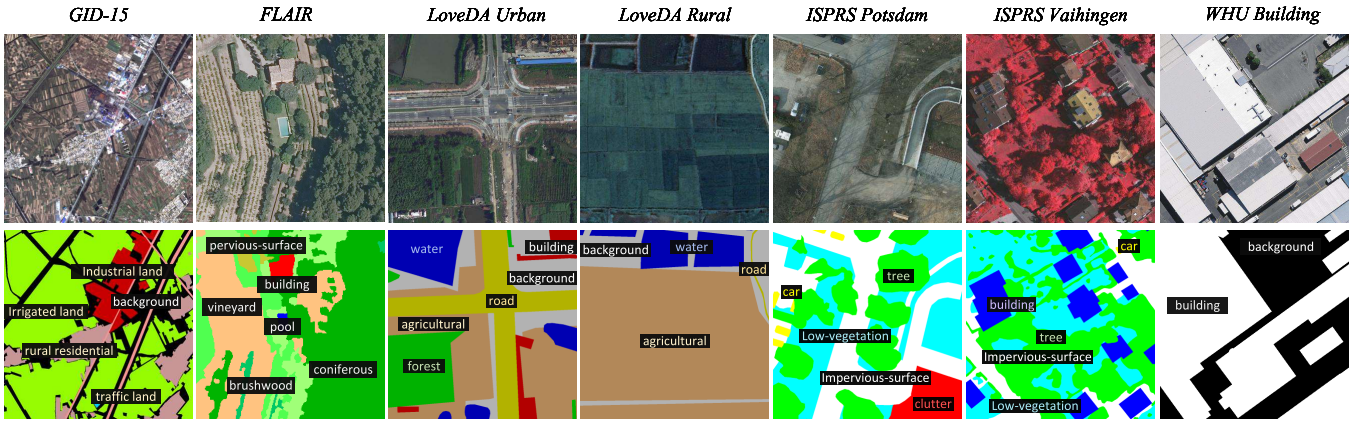


Fig. 5. Typical image samples in the experiments.

TABLE I
LAND COVER CATEGORIES IN DIFFERENT DATASETS

Dataset	Categories
GID-15	paddy field, irrigated land, dry cropland, garden land, arbor forest, shrub land, natural meadow, artificial meadow, industrial land, urban residential, rural residential, traffic land, river, lake, pond, background
FLAIR	building, pervious surface, impervious surface, bare soil, water, coniferous, deciduous, brushwood, snow, vineyard, herbaceous vegetation, agricultural land, plowed land, swimming pool, clear cut, ligneous, mixed, greenhouse, other
LoveDA Urban	background, building, road, water, barren, forest, agricultural
LoveDA Rural	background, building, road, water, barren, forest, agricultural
ISPRS Potsdam	impervious surfaces, building, tree, car, clutter, low vegetation
ISPRS Vaihingen	impervious surfaces, building, tree, car, clutter, low vegetation
WHU Building	building, background

capability emerges from the inherent vision-language alignment learned by CLIP and is further enhanced at the pixel level through the cost aggregation framework.

IV. EXPERIMENTS

A. Datasets

In the experiments, six widely used open-source remote sensing semantic segmentation datasets are employed to evaluate the effectiveness of the proposed approach on OVSS. Their specific categories of different datasets are detailed in Table I. Typical samples are illustrated in Fig. 5, which shows the serious statistical and class heterogeneity of these datasets.

1) GID-15: The Gaofen Image Dataset (GID) [11] contains high-resolution imagery captured by the Gaofen-2 satellite, supporting large-scale remote sensing land cover applications. It includes two subsets: GID-5, intended for broad land cover classification, and GID-15, offering more granular and detailed annotations. The GID-15 dataset contains 150 images with a size of 6800×7200 pixels and is annotated at the pixel level by experts in remote sensing interpretation. Each image is cropped into patches of 1024×1024 pixels, with 4200 used for training and 420 for testing.

2) FLAIR: The FLAIR Dataset [58] is a large-scale land cover segmentation dataset developed by French National Institute of Geographical and Forest Information. It contains very high-resolution (0.2 m) aerial imagery covering about 817 km^2 across 50 regions in metropolitan France, collected between 2018 and 2021. The dataset provides pixel-level annotations for 19 land cover classes such as buildings, vegetation, water, and farmland. With over 20 billion labeled pixels, it exhibits strong spatiotemporal variations across bioclimatic regions and time periods. All images are resized to 512×512 pixels, with 11 925 images for training and 3775 for validation, following the official split.

3) LoveDA Urban and Rural: LoveDA [59] is a land cover semantic segmentation dataset designed for UDA and cross-domain semantic segmentation. It contains two subsets covering urban and rural areas with 5987 high-resolution (1024×1024 pixels) images at 0.3-m spatial resolution from three different cities. The images in this dataset provide three spectral bands—red, green, and blue (RGB)—and cover seven land cover categories: background, building, road, water, barren, forest, and agriculture. The data were collected from three cities in China: Nanjing, Changzhou, and Wuhan. For the Rural scene, the dataset includes 2358 images, split into 1366 training images and 992 testing images. In comparison, the Urban scene consists of 1833 images, with 1156 used for training and 677 for testing.

4) ISPRS Potsdam: The Potsdam dataset is a widely used urban scene semantic segmentation benchmark, consisting of high-resolution RGB orthophotos with a spatial resolution of 0.05 m. It contains 24 true orthophotos, each with a size of 6000 $\times$ 6000 pixels. The dataset provides four spectral bands—infrared, red, green, and blue (IRRGB)—which offer rich spectral information for semantic segmentation tasks. Ground truth annotations include six land cover classes: building, tree, low vegetation, car, impervious surface, and clutter. In our experiments, the dataset is divided into 18 images for

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TABLE II
QUANTITATIVE COMPARISONS OF DIFFERENT MODELS TRAINED ON FLAIR AND GID-15 (%)

Methods	LoveDA Urban				LoveDA Rural				ISPRS Potsdam				ISPRS Vaihingen				WHU Building			
	mIoU	fwIoU	mACC	pACC	mIoU	fwIoU	mACC	pACC	mIoU	fwIoU	mACC	pACC	mIoU	fwIoU	mACC	pACC	mIoU	fwIoU	mACC	pACC
Train on GID-15																				
MaskCLIP [53]	10.99	11.16	22.30	22.30	9.82	16.29	23.02	26.92	9.08	12.32	21.20	29.53	8.11	12.44	15.95	24.16	40.91	66.93	50.00	81.81
SAN [27]	19.10	22.00	32.03	39.68	18.29	23.26	32.36	42.61	12.01	16.30	24.18	34.49	9.10	14.50	19.92	30.89	27.10	29.16	64.45	42.82
CAT-Seg [32]	25.12	25.66	35.57	43.19	22.19	33.13	31.13	52.76	27.10	34.22	45.71	46.60	16.54	24.61	27.80	41.25	59.50	76.34	66.94	86.88
TPOV-Seg	**28.76**	**30.29**	**39.37**	**47.26**	**29.73**	**46.82**	**37.54**	**65.04**	**29.29**	**37.50**	**52.37**	**52.49**	**22.29**	**33.34**	**40.20**	**51.85**	**65.15**	**79.70**	**71.62**	**88.99**
Train on FLAIR																				
MaskCLIP [53]	8.86	8.10	24.27	21.80	10.00	14.09	27.17	*27.90*	11.70	16.67	23.21	33.14	9.45	15.77	19.34	32.91	43.76	69.97	63.15	80.81
SAN [27]	29.01	29.07	50.26	47.70	23.56	20.73	48.74	40.92	33.47	51.65	45.01	68.96	23.86	39.31	36.04	59.85	41.47	47.90	75.59	60.35
CAT-Seg [32]	34.23	32.15	58.99	51.75	27.72	26.90	61.65	46.04	37.39	55.45	**48.62**	72.29	**37.04**	**58.17**	**46.37**	72.86	73.04	81.60	89.72	88.74
TPOV-Seg	**40.71**	**37.36**	**63.95**	**57.35**	**31.87**	**27.02**	**62.58**	**47.74**	**38.17**	**56.87**	48.37	**73.43**	36.46	57.79	45.92	**72.89**	**79.07**	**86.03**	**93.66**	**91.77**

training and six images for testing, and each image is cropped into patches of 1024 × 1024 pixels for model training.

5) ISPRS Vaihingen: The Vaihingen dataset, also designed for urban scene semantic segmentation, consists of 16 high-resolution true orthophotos, each with a size of 2500 × 2000 pixels and a ground sampling distance of 9 cm. It provides three spectral bands—infrared, red, and green (IRRG)—which are sufficient for semantic segmentation in urban environments. The dataset shares the same land cover classes as the Potsdam dataset, including building, tree, low vegetation, car, impervious surface, and clutter. For our experiments, the dataset is divided into a training set of ten images and a testing set of six images. Each image is further cropped into patches of 1024 × 1024 pixels.

6) WHU Building: The WHU Building Dataset [60] consists of aerial images focused on building structures, with a spatial resolution of 0.075 m. The dataset covers an area of 450 km^2 in Christchurch, New Zealand and is designed to support building detection and segmentation tasks in urban environments. With a uniform resolution of 512 × 512 pixels, a total of 1259 images are employed for training and 689 for testing in our experimental setup.

B. Implementation Details

Four widely used evaluation metrics are adopted for a comprehensive assessment of OVSS performance. Specifically, mean intersection over union (mIoU) is the most commonly used metric, which computes the average IoU across all classes. Frequency-weighted IoU (fwIoU) accounts for class frequency, providing a more practical evaluation by assigning higher weights to classes with more pixels. Mean accuracy (mACC) and partial accuracy (pACC) measure the classification accuracy at the class and pixel levels, respectively, to assess the model's predictive precision. These metrics are defined as follows:

$$\text{mIoU} = \frac{1}{C}\sum_{c=1}^{C}\frac{\text{TP}_c}{\text{TP}_c + \text{FP}_c + \text{FN}_c} \tag{16}$$

$$\text{fwIoU} = \frac{1}{\sum_{c=1}^{C} N_c}\sum_{c=1}^{C} N_c \cdot \frac{\text{TP}_c}{\text{TP}_c + \text{FP}_c + \text{FN}_c} \tag{17}$$

$$\text{mACC} = \frac{1}{C}\sum_{c=1}^{C}\frac{\text{TP}_c}{\text{TP}_c + \text{FN}_c} \tag{18}$$

$$\text{pACC} = \frac{\sum_{c=1}^{C}\text{TP}_c}{\sum_{c=1}^{C} N_c} \tag{19}$$

where TP_c, FP_c, and FN_c denote the number of true positives, false positives, and false negatives for class c, respectively. Furthermore, N_c denotes the number of pixels belonging to class c.

In TPOV-Seg, we use a CLIP ViT-B/16 version [61] of CLIP, and two iterative units containing TCGA and spatial aggregation are adopted in the aggregator, i.e., $N = 2$ in the aggregator. Following the training strategy in [32], we adopt a dynamic processing pipeline for handling images of arbitrary sizes. During training, images are first resized using the shortest edge policy while maintaining the original aspect ratio. The images are then padded to ensure all dimensions are divisible by 32 pixels, which facilitates the efficient network processing. We employ an AdamW optimizer for mini-batch training, starting with a learning rate of 0.0002 and applying the cosine decay throughout the training process. The pre-trained model undergoes a fine-tuning process consisting of 40 000 update iterations. For data augmentation, we utilize random cropping with a fixed size of 384 × 384 pixels and color augmentation. All the experiments are implemented on an Nvidia 4090D GPU.

C. Quantitative Comparisons

To validate the effectiveness of the proposed method, we conduct comprehensive experiments across multiple datasets, with a particular focus on unseen categories. Specifically, GID-15, FLAIR, LoveDA Urban, and ISPRS Potsdam are employed as training sets, respectively.

As shown in Table II, when trained on the GID-15 dataset, TPOV-Seg demonstrates strong cross-domain generalization ability. Compared with other methods, it consistently achieves better segmentation performance across various unseen target datasets, including both multiclass and binary settings. For example, on LoveDA rural, it notably improves the region-level consistency and classification accuracy, reflected by higher mACC and fwIoU scores. On the WHU building dataset, despite the domain shift and class imbalance, TPOV-Seg achieves 65.15% mIoU and 88.99% pACC, clearly outperforming other methods. These results confirm the TPOV-Seg's robustness in adapting to significant variations in

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TABLE III
QUANTITATIVE COMPARISONS OF DIFFERENT MODELS TRAINED ON LOVEDA URBAN AND ISPRS POTSDAM (%)

Train on LoveDA Urban																
Methods	LoveDA Rural				ISPRS Potsdam				ISPRS Vaihingen				WHU Building			
	mIoU	fwIoU	mACC	pACC	mIoU	fwIoU	mACC	pACC	mIoU	fwIoU	mACC	pACC	mIoU	fwIoU	mACC	pACC
MaskCLIP [53]	39.10	47.40	49.11	65.39	24.35	32.27	43.24	50.42	24.45	35.97	42.49	52.28	42.90	67.77	51.00	83.18
SAN [27]	18.91	25.30	28.59	47.12	20.23	29.39	35.57	36.41	16.86	28.16	22.27	24.47	77.07	85.25	88.61	91.47
CAT-Seg [32]	42.54	52.02	56.20	67.75	17.20	21.93	37.97	36.94	11.95	18.75	27.70	35.53	80.94	87.50	**93.84**	92.78
TPOV-Seg	**45.05**	**52.71**	**57.13**	**69.09**	**31.10**	**42.17**	**49.09**	**55.66**	**27.55**	**43.04**	**42.07**	**61.06**	**83.46**	**89.58**	92.53	**94.21**
Train on ISPRS Potsdam																
Methods	ISPRS Vaihingen				LoveDA Urban				LoveDA Rural				WHU Building			
	mIoU	fwIoU	mACC	pACC	mIoU	fwIoU	mACC	pACC	mIoU	fwIoU	mACC	pACC	mIoU	fwIoU	mACC	pACC
MaskCLIP [53]	31.37	43.66	42.74	59.82	15.46	12.06	37.95	28.87	7.75	4.67	34.72	12.30	42.90	67.77	51.00	83.18
SAN [27]	31.43	42.20	58.43	56.35	16.48	19.44	30.70	33.74	10.86	14.50	24.85	29.96	72.09	80.49	90.97	87.86
CAT-Seg [32]	39.17	53.06	65.24	68.13	33.29	32.53	51.46	50.93	26.35	26.57	42.90	44.72	70.68	79.12	91.11	86.80
TPOV-Seg	**42.98**	**55.88**	**69.35**	**69.86**	**38.72**	**37.52**	**56.81**	**55.63**	**27.14**	**22.12**	**45.96**	**41.20**	**85.11**	**90.51**	**94.99**	**94.71**

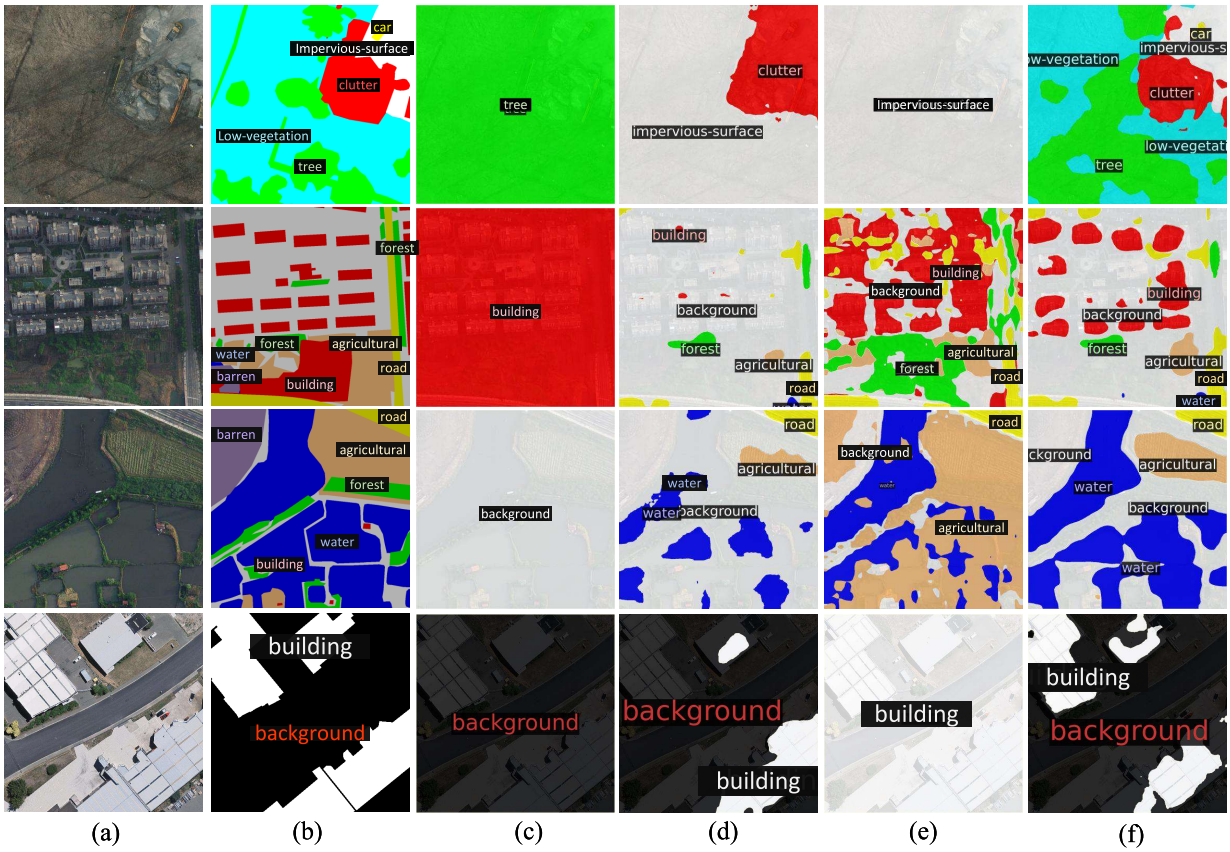


Fig. 6. Visual comparisons of prediction results of different models trained on the GID-15. (a) Input images. (b) Ground truth. (c) MaskCLIP. (d) SAN. (e) CAT-Seg. (f) Ours.

image style, scene layout, and class definitions. When trained on the FLAIR dataset, TPOV-Seg demonstrates strong cross-domain generalization across diverse downstream datasets. Compared with MaskCLIP, SAN, and CAT-Seg, it consistently achieves superior performance in both urban and rural scenes and on high-resolution aerial datasets. Specifically, TPOV-Seg attains 40.71% mIoU and 37.36% fwIoU on LoveDA Urban, outperforming CAT-Seg by 6.48% and 5.21%, respectively, showing the strong adaptability to complex urban textures. It also performs competitively on ISPRS Potsdam and Vaihingen, with 38.17% and 36.46% mIoU, and achieves 79.07% mIoU and 93.66% mACC on WHU Building, clearly surpassing other methods even under class imbalance.

Table III shows the evaluation when the models are trained on LoveDA Urban. TPOV-Seg continues to show competitive or superior performance under open-vocabulary settings. In particular, it performs well on datasets with differing class taxonomies, such as Potsdam and Vaihingen. It surpasses other methods on ISPRS Vaihingen by 3.65% mIoU and 10.71% pACC. This demonstrates its capability to capture transferable semantic knowledge. TPOV-Seg also delivers excellent results on the WHU building dataset, highlighting its flexibility in

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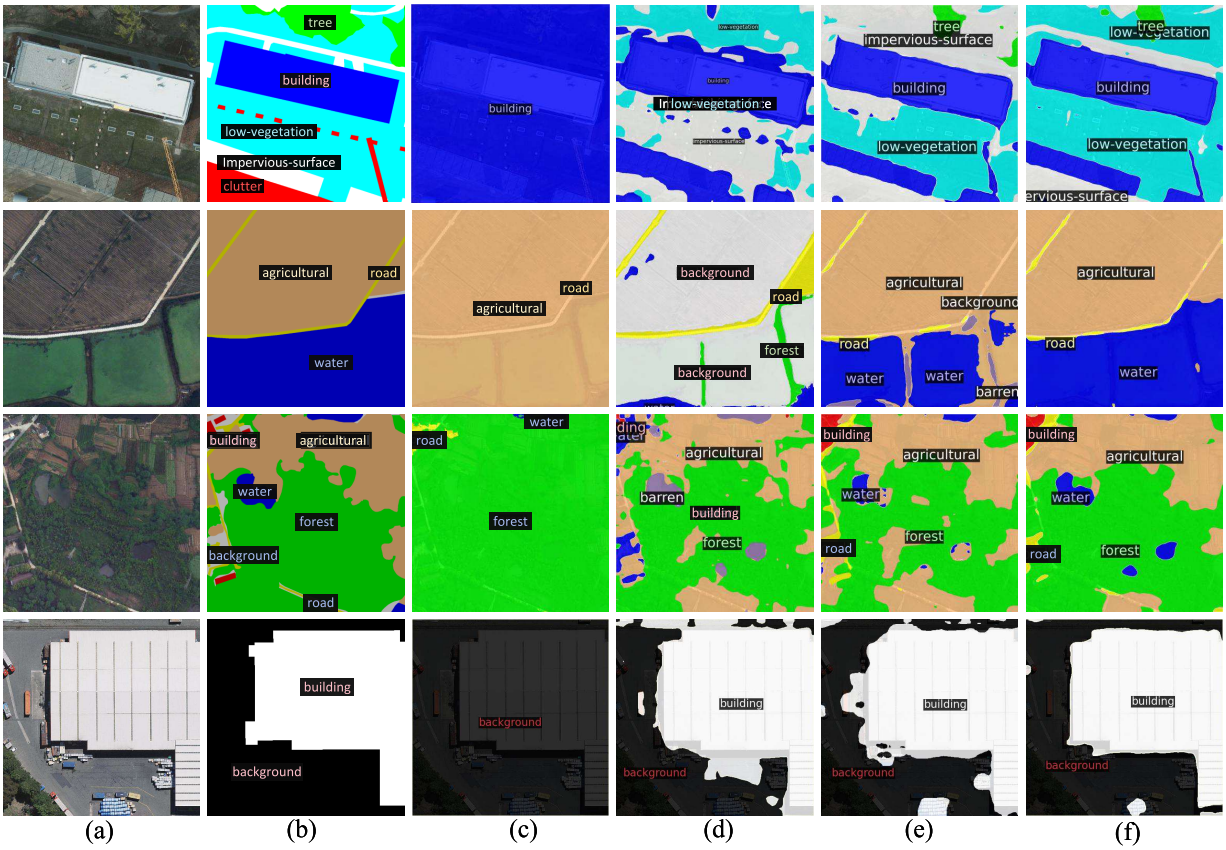


Fig. 7. Visual comparisons of prediction results of different models trained on the FLAIR. (a) Input images. (b) Ground truth. (c) MaskCLIP. (d) SAN. (e) CAT-Seg. (f) Ours.

fine-grained segmentation tasks. The consistently high values in accuracy-based metrics, such as pACC and mIoU, reflect the effectiveness of the TTL-based prompts in guiding cross-domain generalization. When trained on the ISPRS Potsdam dataset, TPOV-Seg maintains its strong generalization across diverse domains. On LoveDA Urban and Rural, which contain different semantic categories and scene characteristics, TPOV-Seg achieves clear gains in mIoU and mACC, with improvements up to 5.45% and 4.75%, respectively. This confirms that the model is not limited by the predefined category structure and can generalize effectively to new domains. Furthermore, its high performance on the WHU building benchmark again illustrates its adaptability across binary and multiclass segmentation settings.

D. Visual Comparison

As shown in Fig. 6, our method demonstrates consistently superior segmentation performance when training on the GID-15 dataset. Our model accurately identifies unseen categories like *low vegetation* and *clutter*, while baseline models like MaskCLIP and SAN misclassify *clutter* as *impervious surface* or fail to capture detailed boundaries. MaskCLIP tends to over-predict background regions as *forests* and SAN suffers from high confusion in urban regions, especially mislabeling building and road pixels. CAT-Seg, while performing relatively better, still produces coarse segmentation boundaries and misidentifies fine-grained classes, such as *road* and *forest*. In the third row, where the scene contains dense spatial patterns and small-scale objects, our method precisely recognizes *water*, *forest*, and *building*, offering a more coherent and structurally correct segmentation map. Compared to CAT-Seg and SAN, which exhibit fragmented or over-smoothed predictions, and MaskCLIP, which struggles with generalization to unseen domain samples, our model achieves much cleaner and boundary-preserving results, validating its strong cross-domain generalization ability.

For qualitative analysis of results trained on the FLAIR data, our method demonstrates superior cross-dataset generalization performance, as shown in Fig. 7. It accurately segments diverse terrain types such as *building*, *tree*, *low vegetation*, and *impervious surface*, while baseline models like MaskCLIP and SAN suffer from noise, fragmentation, and misclassification. CAT-Seg performs better but still lacks boundary precision. Our approach maintains clear distinctions among *agricultural*, *water*, and *road* regions, and delivers structurally consistent and semantically accurate results even in complex spatial scenes. In particular, it achieves precise *building* segmentation with clean boundaries, outperforming all baselines and showing strong generalization on unseen datasets.

Fig. 8 presents results from models trained on the LoveDA Urban dataset. Our model exhibits superior urban semantic understanding across diverse scenarios in the ISPRS Potsdam and Vaihingen datasets. The first two scenes demonstrate our

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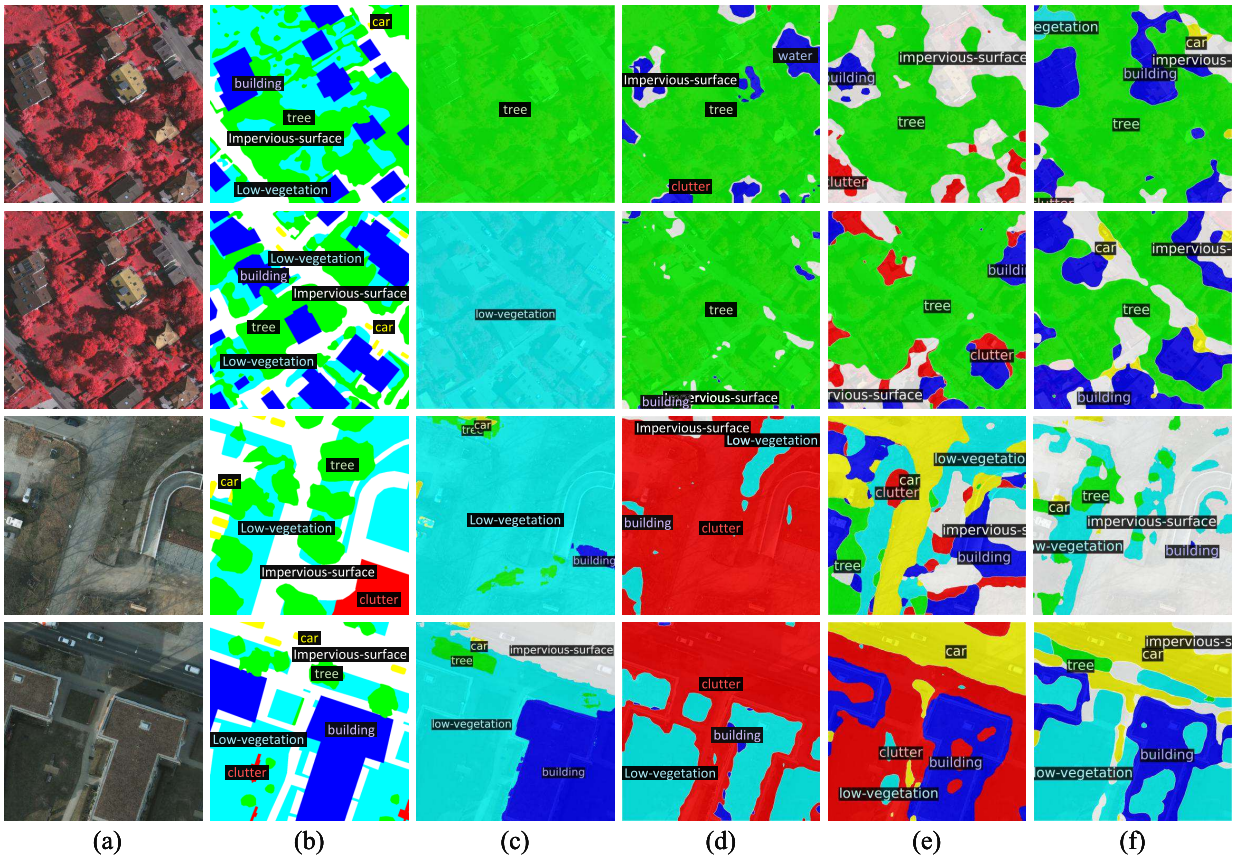


Fig. 8. Visual comparisons of prediction results of different models trained on the LoveDA Urban. (a) Input images. (b) Ground truth. (c) MaskCLIP. (d) SAN. (e) CAT-Seg. (f) Ours.

method's classification ability, particularly in separating *car*, *buildings*, and *impervious surface* from trees with cleaner boundaries than comparative approaches. As shown in the third row, Cat-Seg produces a confused classification result with mixed and unrelated categories. In contrast, our model successfully extracts most of the *low vegetation* and *impervious surface* while effectively suppressing noise. In the fourth row, our model excels in building boundary delineation and clutter recognition, achieving the segmentation precision that clearly surpasses other methods. While CAT-Seg delivers acceptable results in certain regions, MaskCLIP and SAN show notable failures in both object identification and boundary localization.

As shown in Fig. 9, when trained on the ISPRS Potsdam dataset, our model demonstrates robust segmentation performance. It achieves comparable results with CAT-Seg in *barren land* recognition, while outperforming MaskCLIP and SAN in distinguishing *barren* from *agricultural* areas. Our method provides more precise segmentation in *agricultural*, *water*, and *forest* regions and shows clear advantages in complex urban scenes, achieving accurate and boundary-preserving *building* extraction. Overall, the results highlight our model's superior capability in capturing fine-grained urban features and complex land cover patterns, especially for challenging categories, such as *barren land* and clustered buildings.

From the experiments, we can observe that the cross-domain generalization ability of TPOV-Seg is affected by both category heterogeneity and spectral feature discrepancies. The results show that performance degradation is more strongly driven by differences in category definitions and semantic granularity across datasets, while spectral feature discrepancies also contribute, but to a lesser degree. These findings suggest that enhancing adaptive textual representations is crucial for alleviating the effects of category heterogeneity and, in turn, improving robustness in open-world scenarios.

E. Ablation Study

To evaluate the effectiveness of each component in the proposed approach, we conduct a comprehensive ablation study on the Potsdam and Vaihingen validation sets, as shown in Table IV. With all components included, the model achieves the best overall performance. When TGCA is removed while keeping TTL and LTP-Tuning, the mIoU on Potsdam drop to 23.23%, a decline of 7.87%, and pACC drop to 5.05%. Similarly, on Vaihingen, mIoU drop to 2.20%, confirming that this component significantly enhances channel aggregation. Removing LTP-Tuning and TTL instead, the model achieves 30.30% mIoU and 51.24% pACC on Potsdam—showing a drop of 0.80% and 4.42%, respectively. On the Vaihingen, mIoU decreases by 1.98%, and pACC drops by 4.75%. These results indicate that LTP-Tuning and TTL play a critical role in prompt adaptation and improving the semantic discrimination across domains. Furthermore, when only TTL is used and both LTP-Tuning and TGCA are removed, performance drops

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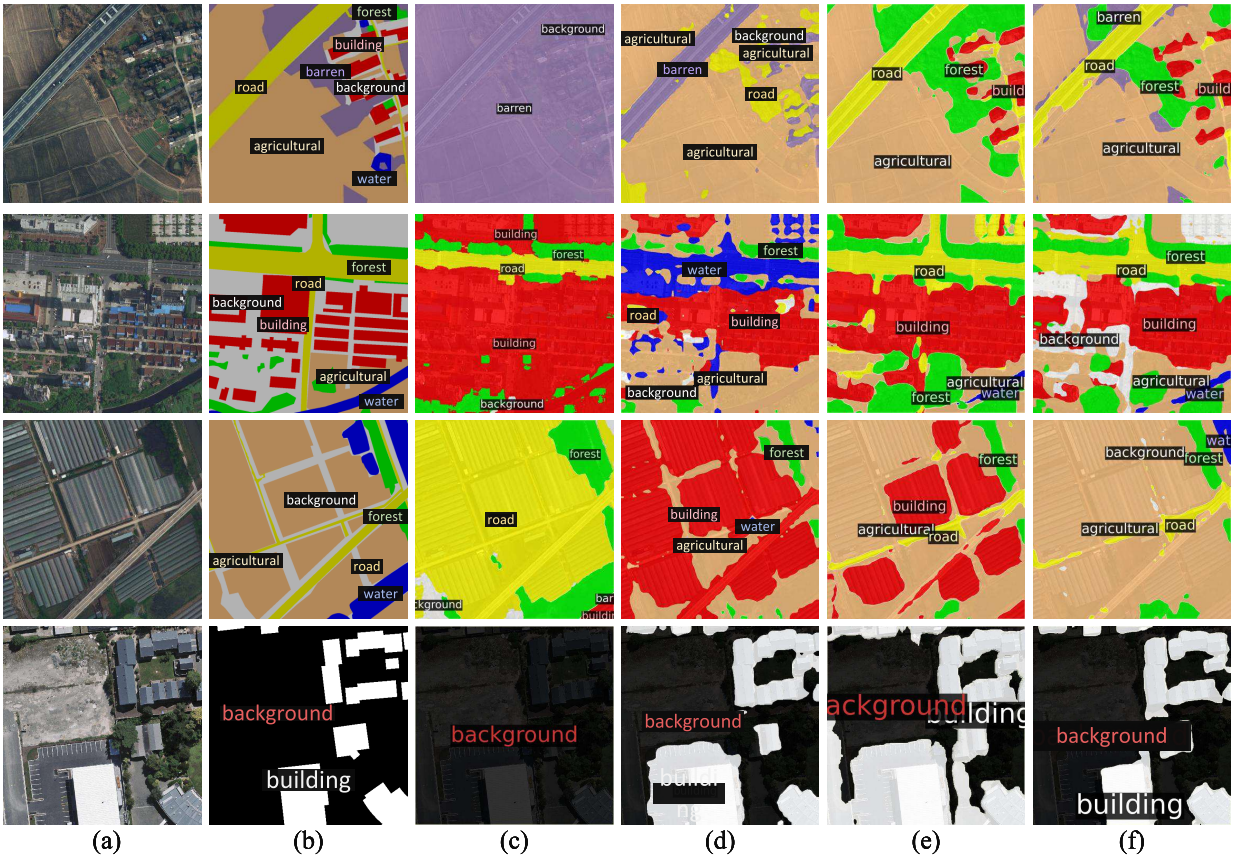


Fig. 9. Visual comparisons of prediction results of different models trained on the ISPRS Potsdam. (a) Input images. (b) Ground truth. (c) MaskCLIP. (d) SAN. (e) CAT-Seg. (f) Ours.

TABLE IV
ABLATION STUDY OF THE KEY COMPONENTS ON THE POTSDAM AND VAIHINGEN VALIDATION SETS

TTL	LTP-Tuning	TGCA	ISPRS Potsdam				ISPRS Vaihingen			
			mIoU	fwIoU	mACC	pACC	mIoU	fwIoU	mACC	pACC
✗	✗	✗	17.20	21.93	37.97	36.94	11.95	18.75	27.70	35.53
✓	✗	✗	22.14	31.60	38.90	45.21	20.30	32.82	33.88	49.83
✗	✓	✗	23.04	34.39	35.47	49.90	23.69	38.56	34.86	55.94
✗	✗	✓	30.30	41.22	48.65	51.24	25.57	40.03	40.18	56.31
✓	✓	✗	23.23	34.26	36.31	50.61	25.35	39.30	36.31	54.21
✓	✗	✓	30.93	39.46	48.27	55.15	26.89	42.05	41.93	58.41
✗	✓	✓	28.94	40.95	45.41	57.94	25.62	41.64	40.01	59.75
✓	✓	✓	**31.10**	**42.17**	**49.09**	**55.66**	**27.55**	**43.04**	**42.07**	**61.06**

drastically. This illustrates that while TTL provides basic textual guidance, it alone is insufficient for handling complex domain shifts without LTP-Tuning and TGCA. These results clearly demonstrate the necessity and complementarity of the three modules in achieving the strong generalization performance in open-vocabulary remote sensing scenarios.

Notably, without TTL, integrating LTP-Tuning with TGCA leads to a decrease in accuracy compared with the case without prompt tuning. The reason is that although LTP-Tuning models contextual relationships, the adapted text embeddings become highly specialized yet lack semantic diversity, which weakens their effectiveness as guidance signals for TGCA's channel-aware aggregation. In contrast, TTL introduces diverse textual representations through synonym expansion and template enhancement, encouraging LTP-Tuning to learn more general semantic features rather than task-specific adaptations. This collaboration enables TGCA to better exploit rich textual guidance and achieve superior cross-domain generalization performance.

F. Category-Level Analysis

We further analyze the categorywise differences observed in the semantic segmentation maps under three domain transfer scenarios, including FLAIR to LoveDA Urban and LoveDA Urban to ISPRS Potsdam.

As shown in Table V, TPOV-Seg improves performance in *agricultural*, *barren*, *forest*, *road*, and *water* classes, showing a

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TABLE V
CATEGORY-LEVEL COMPARISON FROM FLAIR TO LOVEDA URBAN

Methods	Agricultural		Background*		Barren*		Building		Forest*		Road*		Water		Total	
	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	mIoU	pACC
MaskCLIP [53]	0.00	0.00	**6.58**	**10.74**	1.62	1.66	20.15	79.27	8.10	9.70	2.68	2.98	22.93	65.60	8.86	21.81
SAN [27]	43.72	66.51	0.17	0.17	0.09	0.10	**43.53**	72.84	38.11	83.34	22.83	68.96	54.66	59.87	29.01	47.70
CAT-Seg [32]	46.54	54.31	0.19	0.19	16.02	16.64	39.12	92.20	38.40	**85.90**	37.14	**82.27**	62.19	**81.42**	34.23	51.75
TPOV-Seg	**54.71**	**72.77**	0.24	0.25	**30.97**	**35.28**	43.18	**93.85**	**44.48**	85.18	**44.97**	79.60	**66.45**	80.71	**40.71**	**57.35**

* denotes unseen classes.

TABLE VI
CATEGORY-LEVEL COMPARISON FROM LOVEDA URBAN TO ISPRS POTSDAM

Methods	Impervious Surface*		Building		Tree*		Low Vegetation*		Car*		Clutter*		Total	
	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	mIoU	pACC
MaskCLIP [53]	18.86	20.83	62.83	86.45	11.84	15.15	**35.46**	**82.32**	**17.10**	54.21	0.00	0.00	24.35	50.42
SAN [27]	12.71	14.87	76.02	84.05	4.99	5.03	18.37	21.60	2.13	7.48	7.19	**80.39**	20.24	36.41
CAT-Seg [32]	4.45	6.01	52.42	**97.26**	16.88	18.57	16.32	19.15	6.52	**69.26**	6.60	17.55	17.20	36.94
TPOV-Seg	**41.73**	**66.37**	**69.10**	78.12	**23.56**	**26.10**	31.53	38.71	11.26	56.29	**9.43**	28.94	**31.01**	**55.66**

* denotes unseen classes.

balanced segmentation capability across diverse urban scenes. Compared with CAT-Seg and SAN, which exhibit strong results in specific categories, TPOV-Seg demonstrates more stable and comprehensive adaptation performance on unseen classes. In contrast, MaskCLIP shows limited generalization, with low IoU and accuracy across most categories. These results highlight the effectiveness of TPOV-Seg in handling category discrepancies between training and testing datasets. In Table VI, for the LoveDA Urban to ISPRS Potsdam transfer, TPOV-Seg maintains strong adaptability. It achieves the best results in most categories, including *impervious surface*, *tree*, and *clutter*, and maintains competitive performance in *building* and *low vegetation*, indicating a strong capacity to generalize to unseen urban scenes. Notably, although MaskCLIP performs relatively well on *building* and *low vegetation*, it fails to detect *clutter* entirely. SAN and CAT-Seg show limitations in both category diversity and segmentation accuracy. As shown in Tables V and VI, TPOV-Seg does not achieve the best performance for all categories. We attribute this to two main factors: 1) categories with high intraclass variability or ambiguous semantic boundaries (e.g., *background* and *barren*) are more sensitive to domain shifts, which limits precise alignment between textual and visual features and 2) classes with sparse or imbalanced samples (e.g., *car* and *low vegetation*) may bias the model toward dominant categories in cross-domain transfer. Nevertheless, TPOV-Seg consistently achieves superior overall mIoU and pACC, indicating stronger cross-domain generalization despite occasional category-level fluctuations.

G. Robustness Analysis

As shown in Fig. 10, our method demonstrates strong capabilities in both road detection and annotation error identification. In the top row example, our method successfully detected a road that was missing in the original dataset annotations, highlighting our model's ability to identify overlooked features. In the bottom row example, the agricultural area at the top was mislabeled as forest, while the forest area on the right was mislabeled as agricultural land. In both cases, our method provides more precise predictions. This further demonstrates our method's robust semantic understanding capabilities and resilience to annotation noise, showing its effectiveness in both detecting missing linear features and fine-grained land cover classification.

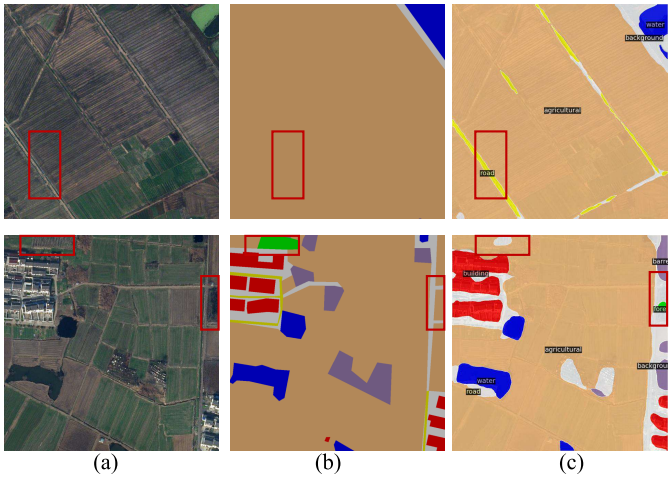


Fig. 10. Performance of TPOV-Seg in high-uncertain areas. (a) Input images. (b) Ground truth. (c) Model predictions. The rectangles highlight the land cover details detected by the model but missing from the ground-truth annotations.

H. Analysis of Prompt Tuning

Different fine-tuning strategies are compared in Table VII, including tuning on image or text branches individually, or both simultaneously. The results show that tuning only the textual branch yields the best performance on both LoveDA Urban and ISPRS Potsdam datasets, while maintaining a comparable number of parameters and inference time. This demonstrates that adapting textual representations effectively enhances cross-modal alignment without overfitting. Furthermore, we investigate the influence of prompt vector length.

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TABLE VII
ANALYSIS OF CLIP TUNING ON THE FLAIR TRAINING SET

No.	Model Variants	LoveDA Urban	ISPRS Potsdam	#param. (M)	Infer Time (10^{-3} s/iter)
Prompt-Tuning Scope for CLIP					
#01	w/o LTP-Tuning	37.31	37.61	155.755	30.9
#02	w/ LTP-Tuning (Img.)	37.66	37.38	155.794	32.3
#03	w/ LTP-Tuning (Txt.)	**40.71**	**38.24**	155.770	33.8
#04	w/ LTP-Tuning (Both)	25.23	22.12	155.821	37.8
Prompt Vector Length in CLIP					
#05	Prompt Len.=2	**40.71**	38.17	155.770	31.6
#06	Prompt Len.=4	37.94	**38.24**	155.783	33.8
#07	Prompt Len.=6	35.69	37.53	155.794	39.4
#08	Prompt Len.=8	23.68	33.32	155.806	40.0
Text Templates in CLIP					
#09	w/o TTL	39.77	37.65	155.770	30.1
#10	w/ TTL by GPT-4.5	40.42	37.88	155.770	49.4
#11	w/ TTL after refinement	**40.71**	**38.24**	155.770	33.8

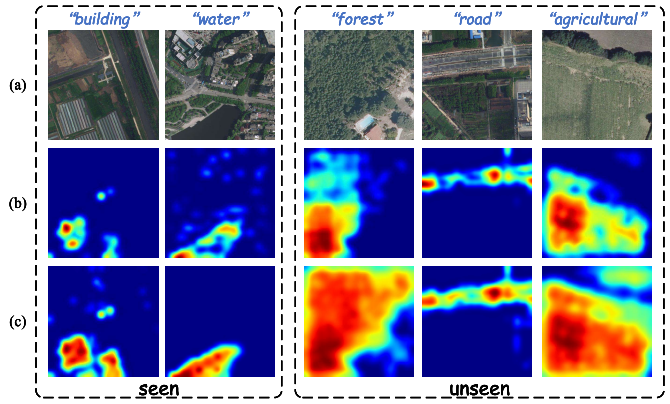


Fig. 11. We visualize the class-discriminative regions that contribute most to the model's prediction based on Grad-CAM. Objects with varying scales from both seen and unseen categories are analyzed. This helps to illustrate how the attention focus shifts before and after applying TGCA. (a) Input image, (b) activation map before TGCA, and (c) activation map after TGCA.

A shorter prompt (length = 2) yields the best performance on the LoveDA Urban dataset, while a moderate length (length = 4) achieves the highest accuracy on the ISPRS Potsdam dataset. In contrast, excessively long prompts lead to performance degradation, suggesting that overly complex textual embeddings introduce noise rather than enhancing the representation quality. To further assess the contribution of the TTL component, we compare three variants: without TTL, with TTL generated by GPT-4.5, and with manually refined TTL. As shown in Table VII, incorporating GPT-4.5-generated TTL slightly improves segmentation performance on both datasets, demonstrating the benefit of introducing diverse textual cues. Further refinement of TTL yields the highest accuracy, confirming that manual filtering enhances the quality and relevance of textual prompts.

I. Model Interpretability Visualization

As illustrated in Fig. 11, Grad-CAM is employed to visualize the activation regions of feature maps by tracing the gradient flow. It can be observed that the cost volumes input to the TGCA module fail to capture fine-grained details and lack sufficient semantic context under different text prompts, particularly for unseen categories and small-scale objects. In contrast, the TGCA-enhanced features demonstrate stronger category-aware discriminative ability for both seen and unseen classes while maintaining robustness across objects of varying scales. This improvement benefits from the effective cross-modal feature aggregation at both the channel and class levels, which allows TGCA to better align visual and textual cues across diverse semantic and spatial conditions.

V. CONCLUSION

In this article, we propose TPOV-Seg, a novel approach based on textually enhanced prompt tuning of VLMs. The core idea is to introduce rich and diverse textual information to enhance the semantic understanding and recognition capabilities of VLMs for segmenting unseen categories. Specifically, a remote sensing-specific TTL is proposed, which leverages synonymous vocabulary combinations to enrich textual representations and enhance semantic understanding across domains. In addition, a LTP-Tuning strategy is introduced to adapt the text encoder to remote sensing characteristics through contextual modeling of learnable word embeddings, while keeping the pretrained parameters frozen. In addition, a TGCA is developed to facilitate the class and spatial aggregation by first enhancing informative feature exchange across channel dimensions. Guided by TTL-enhanced textual features, this mechanism strengthens discriminative semantic features and suppresses redundant channels, enabling more effective modeling of semantic relationships and spatial structures. Collectively, these components provide the framework with superior generalization capabilities for OVSS tasks.

Extensive experiments on five widely used remote sensing semantic segmentation datasets demonstrated that our approach achieved superior OVSS performance. In addition, the qualitative analysis demonstrates that our method exhibits stronger recognition capabilities for unseen categories. Ablation studies verify the effectiveness of the proposed textual enhancement modules, which enrich semantic representations and enhance the model's ability to capture spatial structures and discriminative features through an efficient prompt tuning strategy. The limitation lies in that the model's segmentation performance on blurry boundaries in highly complex remote sensing scenes still requires improvement. In future work, we plan to enhance the model's segmentation capabilities by introducing edge priors. Furthermore, we will incorporate multimodal image data, such as hyperspectral and SAR imagery, to further extend the framework's applicability.

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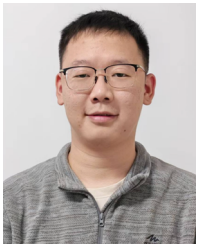
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